SMT2020—A Semiconductor Manufacturing Testbed

Denny Kopp, Michael Hassoun, Adar Kalir, and Lars Mönch

***Abstract*—We present a new set of simulation models, organized in a testbed. The aim of the testbed consists in providing researchers with a platform able to credibly represent the complexity of modern semiconductor manufacturing. The testbed is open to public use, and include so far four models. A high-volume/low-mix model and a low-volume/high-mix model constitutes the foundation of the testbed. Two additional models incorporate the complexity of engineering lots. We conclude this paper by presenting a case study that demonstrates that the third and fourth model can be used to assess the performance of integrated dispatching strategies for production and engineering lots.**

***Index Terms*—Discrete-event simulation, semiconductor manufacturing, testbed, production planning and control, simulation experiments.**

I. Introduction

MANY semiconductor companies rely on simulation as an important tool for decision support [1]. Moreover, discrete-event simulation equips researchers in semiconductor manufacturing with a real-world like, risk-free environment [2] to test policies. The standard models used by researchers, are usually made available to the community as datasets, each representing a functioning fab, and collected together under a testing platform often referred as a testbed. A testbed also includes a documentation of the features of the different datasets together with some benchmark results for replication purposes. Simulation models designed and applied by practitioners usually tend to represent as much as possible the semiconductor wafer fabrication facility (wafer fab) at hand, i.e., high-fidelity models are used. It is possible to study long-term phenomena in wafer fabs and semiconductor supply chains using multiple scenarios and replications. Tactical and also time-critical operational decisions can be made using simulation models that are initialized based on the current situation in the manufacturing system. Various full factory simulation studies are conducted using specific simulation settings, e.g., on production planning, and on factory performance in a transient period [3], [4]. These studies provide often useful insights but are somehow limited in their abilities to share, replicate, and compare results with other researchers and practitioners using simulation. Academic researchers are interested in using generic simulation models to study phenomena and design policies at the different decision levels of a wafer fab. But designing and building simulation models for the purpose of a unique research team leads to the following disadvantages:

1. It is likely that the obtained simulation results and insights are strongly model-dependent.
2. Ensuring reproducibility of the simulation results by other researchers is extremely difficult or simply impossible.
3. The conclusions reached on simplified simulation models used by academic researchers are often received with suspicion by practitioners in wafer fabs. Since a critical level of complexity is missing the results are often considered as not relevant for actual fab operations.

Due to these reasons, a standardized simulation testbed that addresses the following two needs is highly desirable:

1. The simulation models of the testbed serve as a common testing platform for researchers.
2. The models serve as a bridge between academic research and industrial application of simulation.

The first of these needs is the easier one to achieve since all that is required is a level of details and plasticity that fits the need of researchers. The second need, however, is more difficult to reach. It requires that model details, features, and scale satisfy practitioners if the conclusions supported by the simulation models of the testbed are to be accepted as relevant to the industry. Among many wafer fab simulation models, the Measurement and Improvement of Manufacturing Capacity (MIMAC) datasets [5] have been highly successful over the last 25 years. Various research efforts are based on the simulation models of this testbed. These models form until today the best available testbed for simulation-based semiconductor manufacturing research, and, with no

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better alternative, remain the major benchmark of choice for researchers, e.g., [6], [7]. However, while some of the model features are outdated, others are lacking when modern wafer fabs are considered.

Therefore, it is a highly desirable undertaking to propose and create a new set of simulation models in the same spirit but with features that are more appropriate for representing state-of-the-art wafer fabs. Recently, the main features of such a testbed were formulated by Hassoun and Kalir [8]. We refer to the testbed as Semiconductor Manufacturing Testbed 2020, or SMT2020 for short. A high-volume/low-mix simulation model as part of SMT2020 is already described by the present authors in [9]. The present paper is a considerably extended version of [9]. The model presented in [9] is improved and three additional simulation models are presented which are representative of the current technology intricacies. Moreover, an application of the SMT2020 simulation models is outlined by means of a case study.

The remainder of this paper is organized as follows. Requirements for a semiconductor manufacturing simulation testbed are formulated in Section II, including a discussion of previous work. The SMT2020 testbed features are described in Section III. The performance of the different simulation models is studied in Section IV. An application of the SMT2020 simulation models is discussed by means of a case study in Section V. The usage of the testbed is described in Section VI. Conclusions and future research directions are presented in Section VII.

II. Problem Analysis and Related Work

A. Requirements for a State-of-the-Art Simulation Testbed

We briefly summarize the main functional requirements for the SMT2020 testbed based on [8], [9]:

1. *Support of Different Simulation Model Types:* The testbed has to include simulation models of low-volume/high-mix (LV/HM) and high-volume/low-mix (HV/LM) wafer fabs. The LV/HM setting is used to represent make-to-order (MTO) environments such as foundries, producing either logic or memory devices. The corresponding models have to contain at least ten products. Due dates are required for all lots, the moving entities in wafer fabs. A lot consists of wafers, thin silicon discs. In the HV/LM setting, two high-volume products should be modeled to represent dedicated fabs, typically producing logic devices in a make-to-stock (MTS) environment. The number of process steps per route should be more than 500 to reflect the complexity of modern wafer fabs [1]. More than 100 tool groups, i.e., work centers, have to be modeled in each simulation model for the same reason. Moreover, given the importance of engineering activities in wafer fabs [10], SMT2020 has to include simulation models that incorporate engineering activities.
2. *Availability Modeling:* Scheduled and unscheduled downtimes should be represented in all simulations models of SMT2020 since preventative maintenance (PM) actions and tool breakdowns are typical for the complicated machinery of wafer fabs [1].
3. *Machine Rate Modeling:* Batch processing machines, i.e. tools where several lots are processed at the same time on a single tool [1] and cascading tools [11] have to be represented in the models to mimic important process features. Load and unload times on the tools and transportation times between tool groups have to be modeled. Moreover, setups, most importantly sequence-dependent setups which are often longer than the pure processing time [1], [2], have to be modeled in some detail to represent the offered capacity in a correct way.
4. *Functional Area Specific Complexities:* Important features of certain functional areas have to be represented in SMT2020 models. Such features are, for instance, lot-to-lens dedications for steppers in the photolithography area, sampling rates for metrology tools, and critical queue time (CQT) constraints for the dry etch area.
5. *Operational Complexities:* High priority lots, so-called hot lots, are known to have a disruptive impact on the work-in-process (WIP) flow [12], and as such, are to be represented in all SMT2020 models. It is important to represent engineering lots in some simulation models of the testbed due to the fact that up to one third of all WIP lots in a wafer fab can belong to engineering activities [10]. CQT process segments [13] are another disruptive feature of many modern fabs and these have been incorporated at least in some of the SMT2020 models. In contrast to the MIMAC datasets [5], [14], there is no need to model operators given the degree of automation with respect to loading, unloading, and lot transport in modern wafer fabs.

We identify two additional non-functional requirements related to the usage of the SMT2020 testbed:

1. *Representation of the Simulation Models of the Testbed:* The simulation models belonging to the SMT2020 testbed should be represented in such a way that potential users can implement them without relying on a proprietary simulation tool. ASCII or Excel files with a format similar to the MIMAC datasets [14] and their derivatives [15] are preferable.
2. *Providing Performance Measure Values of the Simulation Models:* The values of common performance measures in wafer fabs including cycle time (CT), availability per functional area, tool utilization, bottlenecks, and operating curves must be reported to provide potential users with a way to check whether their implementation of a simulation model works as intended or not.

B. Discussion of Related Work

Discrete-event simulation is an important tool for decision-making in wafer fabs [1]. There is some evidence that building a full wafer fab simulation model from scratch is time-consuming and error-prone [2]. Therefore, researchers in semiconductor manufacturing are interested in applying simulation models of testbeds. The following four simulation testbeds are fairly often applied by researchers.

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1. The most common one is the MIMAC testbed [5], [14]. The testbed is publicly available under [16] and is composed of six full fab models of different complexity. Each wafer fab model includes up to 260 machines used to process several products (up to 21) whose route length can reach 340 process steps. One of the most popular models, the MIMAC dataset 1, is used in numerous research efforts. Based on this model, [17] assesses scheduling algorithms for wafer fabs in a rolling horizon setting. In [18], the formation of multiple orders per job and release algorithms is discussed. The MIMAC dataset is also applied to evaluate release schedules for wafer fabs in a dynamic and stochastic setting by executing them in the simulation model [19], [20]. The MIMAC 1 dataset is applied in [21], [22] to evaluate qualification management decisions in a rolling horizon setting. MIMAC dataset 6 is also used extensively in research. For example, dispatching problems are tackled in [23], [24], while genetic programming is used in [25] and [26] to evaluate dispatching rules. In [27], dataset 1 is again used to design and assess methods for CT prediction in wafer fabs. Engineering lots are added to dataset 1 in [28] to assess integrated planning formulations for both production and engineering lots in a rolling horizon setting.
2. The MiniFab model proposed by Spier and Kempf in [29] is a reduced-scale simulation model that mimics the typical behavior of a wafer fab. It contains reentrant process flows, batch processes, and sequence-dependent setup times. Using this model, problem instances for evaluating a full wafer fab scheduling approach are derived in [30]. An integrated scheduling and automated material handling approach is assessed in [31] in a rolling horizon setting using the MiniFab model.
3. A scaled-down representation of a real-world wafer fab at Harris Semiconductor is described in [32]. It is used by Kacar and Uzsoy [33] to estimate clearing functions using simulation-based optimization. Production planning approaches for wafer fabs are assessed in a rolling horizon setting using this model in [34], [35].
4. A generic 300 mm model including automated material handing features is proposed by SEMATECH in [36]. It is used, for instance, in [37] to generate problem instances for global fab-wide scheduling approaches. This model is less frequently applied in comparison to the three other testbeds and simulation models described before.

Table I summarizes the main characteristics of existing testbeds and simulation models. All the discussed testbeds and simulation models except the SEMATECH 300mm model are publicly available at [16].

We see from Table I that the model scale is too small, i.e., the number of machines and work centers is too low, and the process routes are too short to accurately represent state-of-the-art wafer fabs. A process route is a sequence of process steps necessary to the manufacture of a certain product. A set of possible machines is assigned to each step within a route [1]. Moreover, only limited efforts exist to provide testbeds for semiconductor supply chain simulation. One example is the testbed designed by Ewen *et al.* [15]. The nodes of the simulations models that represent wafer fabs are based on the MIMAC dataset 1. A generic data model for semiconductor supply chains as a prerequisite for simulation models is proposed in [38].

TABLE I
PROPERTIES OF TESTBEDS/SIMULATION MODELS

Name	# Machines/ Work centers	#Products	#Process steps
MIMAC datasets	up to 260/up to 85	up to 21	up to 280
MiniFab	5/3	2	6
Harris (Kayton)	12/11	3	up to 22
SEMATECH 300mm	275/103	1	364

Overall, we conclude that the existing set of testbeds and simulation models, despite having provided the research community an excellent platform to tackle a variety of research questions, is now aging. These models are still excellent tools to study operations of legacy wafer fabs, yet the addition of new models that incorporate the scale and complexity of modern wafer fabs is required.

III. SMT2020 FEATURES

We provide details on the various SMT2020 models available at this stage. Building baseline models do not require them to be precisely matching any specific fab, just to provide the community a baseline that makes sense and is widely accepted. During the preparation of this paper, feedback from the community was collected to ensure as much as possible that reasonable features are chosen for the datasets and that they are parameterized in an appropriate way.

The models are designed and tested under steady state. Specific transient scenarios are left to the users to design and run to answer their specific needs. In general, the models presented in SMT2020 can be seen as an update on the MIMAC datasets. The main differences between these two generations of testbeds are the following:

- The larger size (number of tools and steps) proposed in SMT2020 better fits current fab scale.
- SMT2020 provides a detailed modelling of scheduled and unscheduled downtime instead of a unified distribution for stoppage and repair times.
- Hot Lots and engineering lots are represented.
- Complex operational features that were irrelevant or non-existent at the time of the MIMAC creation are represented:
  - lot-to-Lens dedication
  - cascading
  - critical queue times.

A. General Structure

The different datasets in SMT2020 are realistic in their scale. Depending on the models, close to a thousand machines of 105 types are modeled. Identical machines share all their

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characteristics (availability patterns, machine rates, qualifications etc.) and are grouped in what we denote as a tool group. These are grouped in the following functional areas, including three metrology areas:

- diffusion
- wet etch
- dry etch
- implant
- lithography
- dielectric
- planarization
- thin films
- lithography metrology
- thin films metrology
- defectivity metrology.

Because we avoid modeling complex availability and machine rate schemes that allow partial machine functions or hybrid tools such as in dry etch and thin films, individual entities (chambers) are defined as machines (tools).

Ten products with route lengths ranging from 242 to 583 operations are defined. Routes 1 to 3 are long (more than 500 steps) and the rest are shorter. The maximum number of layers is 44. These routes share some of their operations with other routes. Datasets 1 and 3 (HV/LM) are based on two products only (one long and one short), while datasets 2 and 4 (LV/HM) make use of all ten products. Datasets 3 and 4 add engineering lots to datasets 1 and 2, respectively.

The production rates are defined for each tool type based on a processing time for a single wafer, a lot of 25 wafers, or for a batch. Batches are used in the diffusion area and are groups of lots that are processed at the same time on a single diffusion tube. The processing time is sampled from a continuous uniform distribution ($+-5\%$ around the nominal value) in each simulation run. A fixed loading and unloading time is added to the processing time for the lot or batch. The time required to transfer lots between tool groups is represented by a delay uniformly sampled between 5 to 10 minutes.

In all SMT2020 datasets, the maximum batch size is set between 100 and 150 wafers with an allowance of one lot to allow for some flexibility in the operation. For example, if the maximum batch size is 6 lots (150 wafers), a batch can be processed as soon as 5 lots are found waiting in the tool group queue.

A special regime, denoted as cascading, takes place in certain tools that allow different parts, i.e., wafers or lots, to undergo part of their process in parallel with others. This is illustrated in Figure 1 for the simplest case of a two-stage process. Practically, this means that the time required for the first process unit (wafer or lot) of a group is longer than for the remainder. It also means that loading and unloading phases are only accounted for once, at the beginning and end of the cascade. The remaining loading/unloading operations take place in parallel with processing of other lots and therefore do not contribute to the processing time. All tools in implant, planarization, thin films and wet etch areas, and about half of the tools in dielectric area operate with cascading.

When setups apply, a setup avoidance function is added to the regular prioritization scheme that forces the tool to remain

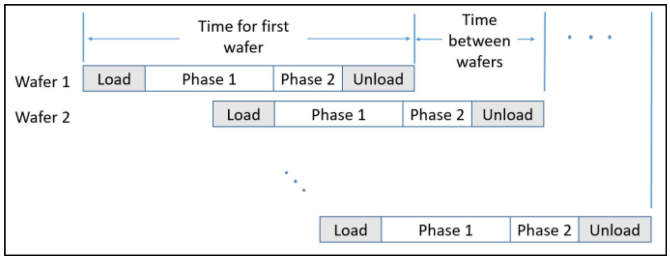


Fig. 1. Wafer cascading.

set on the same operation as long as there are lots waiting for it in the queue. Hot lots are triggered by a change in demand. In that regard, whether it is a direct demand from a customer in a make-to-order system or a sudden depleting of stock in a make-to-stock system makes little difference at the operational level, and the practice shows that there are no across the board differences in the proportion of hot lots in either LV/HM or HV/LM systems. This is why, in all models, the same proportion of 2.5% of all lots have been defined as hot lots which means that they take precedence on all other lots, and when necessary, trigger setups.

For each of the proposed datasets, the number of tools in each tool group was determined to attain a certain target level of utilization coherent with what is found in the industry within the tool specific functional area, and for a loading level of about 10,000 wafer starts per week (WSPW). As is usually the case in fabs, and with the exception of a few chosen tools, we avoid having what is called in the fab jargon "one of a kind" tools, meaning that a minimum number of two machines was set to ensure a minimum tool redundancy.

B. Availability Modeling

The level of detail in the availability modeling sets the models in SMT2020 apart from their predecessors that tend to model all stoppages as a set of distributions for the time between failures and the time to repair. In SMT2020 PM, also known as scheduled down time (SDT) and breakdowns, i.e., unscheduled down time (UDT), are modeled separately. For each area, UDT and SDT are carefully balanced. In areas known for their stability like metrology or planarization, we aim at 90% of the total down time accounted by SDT and only 10% by UDT, while in lithography, where reliability is an issue, the target for SDT is set at 50% of the total down time only.

1) UDT: Under the UDT are found a wide range of events causing production to stop unexpectedly: First and foremost, equipment malfunction, but also SPC out-of-control, either recipe-specific or related to defectivity, delays or breakdowns in ancillary or transport systems, shortages of replacement parts required in troubleshooting either in the fab or from the supplier etc. Although other distributions are sometimes better suited to precisely render the TBF and TTR distribution, the exponential distribution offers a reasonable baseline (see [39]), and is simple to implement. This is why we use an exponential distribution to sample both time before failure (TBF) and time to repair (TTR). The mean time before failure (MTBF) is set

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at one week for all tools. The mean time to repair (MTTR) is then computed to reach a certain target availability and a certain target balance between UDT and SDT, both defined at the area level.

2) SDT: For lithography, diffusion, and all types of metrology, the maintenance is time-based. All tools undergo a long quarterly PM and a monthly PM of medium duration. In addition, lithography tools undergo a shorter weekly PM. Maintenance for all other tools is based on usage (counter-based). To provide a realistic baseline PM frequency for these tools, all counters are set to trigger a long duration PM quarterly, a medium duration PM monthly, and a short duration PM weekly.

C. Metrology and Quality Modeling

The frequency of inspection on metrology tools is defined by a sampling rate set individually for each operation. All lots are inspected (100% rate) for approximately half of lithography metrology operations, the other half is characterized by measurement rates between 10% and 50%. Thin films metrology operations have inspection rates ranging from 30% to 40%. Typically, the process front-end where the transistors are built is more sensitive to particle contamination than the back-end where the metal layers are formed. This is why defectivity metrology uses a sampling rate set between 50% and 60% in the front-end and 10% in the back-end. In some lithography operations (about one third) 0.5% to 2% of lots are reworked. Rework involves a re-doing of the process segment immediately preceding the rework operation, following the same recipes and on the same machines.

D. Constraints

To cope with setups, operators tend to run long series of lots of the same operation on a machine before switching to another. This mode of operation is one of the major sources of variability that plagues WIP flow in fabs. Setups are therefore a critical feature of the proposed models, and are incorporated in three areas. Two tool groups in the lithography area undergo setups for reticle change. In the back-end of the process two dry etch tool groups have simple sequence-dependent setups (two groups of operations). In the implant area, setups are required for gas change, and so three setup groups are defined.

In lithography, a lot-to-lens dedication scheme is applied on three different tool groups, and for three groups of operations. The sensibility to overlay issues is a characteristic of modern processes as opposed to older ones. Depending on the process, the machinery, and the product value, companies differ in their way to tackle this topic. They may simply accept the yield loss if the value is low and the time to market is critical (typically for memories) or try to balance between operational and quality concerns either by running critical steps under a strict Lot-to-Lens dedication scheme or by allowing some kind of tool grouping based on their optical characteristics, allowing lots to be processed indistinctively within said groups. In our models, when a lot-to-lens dedication applies, once a lot is processed on a specific tool for the first of these operations, it must be returned to the same tool for the remaining operations in the group.

CQTs are defined and provided for the datasets, but no special dispatching rule is applied at the entrance of a CQT segment at this point.

E. Specifics of SMT2020 Models

1) Preliminary Steps: One of the key metrics of a fab behavior is CT. Since it is product-dependent, and in order to be able to present aggregated results, we normalize each product's CT by its minimum achievable value, denoted as raw processing time (RPT). The result is what is known as flow factor (FF) [1], calculated as the ratio of the product CT and its RPT. As a preliminary step, we first run a model with lots starting at very long intervals. As a result, these lots progress in an empty fab and their waiting time is close to zero (the specific model is unimportant since the number of tools has no impact on a single lot trajectory). In such a setup, no minimum batch size is required for these lots in order not to wait extremely long period of times at diffusion tools. The products' minimum CT is then measured to obtain the RPT. The source of variability in the resultant CT lays in the fact that some lots experience delays due to PMs or that they are sampled in metrology steps. The resultant RPTs for all ten products (regular and hot lots) are given in Table II.

2) Dataset 1: For this first model which aims at representing a HV/LM fab, only routes 3 and 4 are used, whose lengths are 583 and 343 steps, respectively. Although the capacity of the fab is around 10,200 WSPW, the proposed working point in order to reach a realistic utilization is 10,000 WSPW (5,000 WSPW for each product), from which 2.5% are hot lots. This translates to the introduction of a new regular lot of each product every 51.69 minutes, and of a hot lot every 2,016 minutes. Often, high-volume manufacturing is associated with a MTS mode. Therefore, we do not represent any due date for the lots and adopt a simple First-Come First-Served (FCFS) dispatching rule along the process.

3) Dataset 2: This second model aims at representing a LV/HM fab. All ten products, with route length ranging from 242 steps for product 5 to 583 steps for product 3, are used and launched into the fab at the same rate. Although the model seems stable for a load as high as 10,250 WSPW, we decided to set the working point of this model to 10,000 WSPW (1,000 WSPW per product) to reach a realistic maximum utilization. Here too, 2.5% of lots are hot lots. This translates to the introduction of a new regular lot every 258.46 minutes and of a hot lot every 10,080 minutes for each product.

This basic high-mix model aims at representing a foundry fab which is mostly running in MTO mode. Thus, the critical ratio (CR) dispatching rule is used to prioritize between lots. CR can be considered as a derivative of the earliest due date (EDD) rule that takes into consideration not only the due date but also the remaining process time required to complete production. The CR index of a lot is defined as the ratio of the time remaining to due date and the processing time required to completion [1]. In dataset 2, lot due dates are determined as:

$$d_j := r_j + z_j \cdot FF_i \cdot RPT_i, \tag{1}$$

where d_j and r_j are the due date and release date of lot j, respectively. It is assumed that lot j belongs to product i. The

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TABLE II
BASELINE FLOW FACTORS FOR DATASET 2

		FF	
Product	**RPT (days)**	**Regular lots**	**Hot Lots**
1	23.76	2.10	1.43
2	25.46	2.06	1.42
3	27.18	2.07	1.41
4	15.96	2.02	1.45
5	10.99	2.17	1.54
6	14.16	2.12	1.48
7	16.85	2.08	1.43
8	17.47	2.17	1.46
9	18.27	2.14	1.46
10	18.98	2.11	1.46

quantity z_j is sampled from a continuous uniform distribution between 0.8 and 1.2 to randomize due dates, RPT_i is product *i's* RPT, and FF_i is a user-determined FF for product i. For each product i, baseline values for FF_i were calculated as the 92% percentile FF while running the model under a FCFS rule at 10,000 WSPW. In Table II we provide the RPTs and baseline FF values both for regular and hot lots.

4) Datasets 3 and 4: Datasets 3 and 4 are extensions of datasets 1 and 2 that include engineering products. Dataset 3 represents a HV/LM wafer fab with one engineering product whose route is inherited from route 3. For this dataset, 40 engineering lots are released per week in addition to the 10,000 WSPW, i.e., 400 production lots. The FCFS rule is applied as for dataset 1. Dataset 4 is based on the same ten products as dataset 2 with the addition of three engineering products derived from products 1, 2, and 3. It is a LV/HM wafer fab. Given a release of 400 production lots per week, a total of 80 engineering lots are launched per week, while the number of lots for each engineering product is the same. The due dates are set as described for dataset 2, and the CR rule is applied as for dataset 2.

20% of all engineering lots are hot lots in both datasets. The first and the second half of the engineering lots are released on Monday and Wednesday morning each week, respectively. The number of wafers per engineering lot is distributed according to $DU[1, 10]$ where $DU[a, b]$ denotes a discrete uniform distribution over the integers $\{a, \ldots, b\}$.

The difference between the routes of production and engineering products consists in an additional calibration effort for engineering lots on specific operations. First, a setup has to be performed for each operation of engineering lots on the planarization tools. The setup time is distributed according to $U[15, 60]$, where $U[a, b]$ refers to a continuous uniform distribution over the interval $[a, b]$. An additional calibration time is also required on lithography tools for reticle change, i.e., a calibration time of 16 or 30 minutes is added to the baseline setups time. The overall time for calibration and reticle change in minutes is distributed according to $U[16, 32]$ or $U[30, 60]$ except for the last three photolithography steps of the routes where the overall time is distributed according to $U[52.5, 127.5]$ due to the fact that calibration is more crucial for these operations. Batches can only contain production and engineering lots belonging to the same route and same operation, e.g., production lots belonging to product 3 and engineering lots from the engineering product derived from product 3 can be batched together.

Due to the higher load, more tools than in datasets 1 and 2 are required in datasets 3 and 4, especially for tool groups with lot-based processing. Moreover, the counters for the PMs are changed for both datasets. This is the main reason why we introduce the two new datasets with engineering features.

TABLE III
MAIN CHARACTERISTICS AND PERFORMANCE MEASURES OF DATASET 1

	Population		**Availability**		**Utilization**	
Area	**Number of tool groups**	**Number of tools**	**Average Availability**	**Proportion of SDT**	**Average Utilization**	**Maximum Utilization**
Dielectric	10	59	80.3%	71.0%	85.0%	89.8%
Diffusion	10	75	92.5%	80.3%	84.1%	90.0%
Dry Etch	21	362	88.5%	80.3%	86.0%	91.8%
Implant	9	35	80.4%	71.2%	58.2%	89.9%
Lithography	11	203	86.4%	51.7%	86.6%	97.4%
Planarization	6	34	80.1%	90.1%	71.2%	87.2%
Thin Films	11	90	85.2%	71.0%	82.5%	90.3%
Wet Etch	14	112	89.0%	80.3%	79.4%	89.7%
Defectivity Metrology	7	16	96.5%	90.0%	52.6%	84.0%
Litho Metrology	4	53	96.5%	90.1%	90.1%	91.7%
Thin Films Metrology	2	4	96.5%	90.2%	46.1%	74.9%
Total	105	1043				

IV. PERFORMANCE OF THE DIFFERENT MODELS

A. Availability, Utilization, and Bottlenecks

An overview of the characteristics and performance measures of each functional area of SMT2020 datasets 1 and 2 is presented in Tables III and IV. These are reviewed at the tool group level in the electronic supplement that will be provided online. The first two columns provide the number of tool groups and total number of tools in the area.

Tool availability is the result of a set of decisions about the duration (or frequency) of PMs and the relative proportion of SDT, which is representative of the level of availability randomness and has an important impact on the variability of the tool's capacity. In Tables III and IV we provide the average availability in each area and the proportion of SDT.

Metrology tools show the highest availability and are also more stable in the fab, with only about 10% of their down-time attributed to breakdowns. On the other end of the scale, tools in implant, dielectric, and planarization have availability values of around 80% on average. Lithography tools are the most unstable, with roughly half of the down time attributed to breakdowns.

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TABLE IV
MAIN CHARACTERISTICS AND PERFORMANCE MEASURES OF DATASET 2

	Population		Availability		Utilization	
Area	Number of tool groups	Number of tools	Average Availability	Proportion of SDT	Average Utilization	Maximum Utilization
Dielectric	10	54	80.4%	71.4%	80.7%	89.9%
Diffusion	10	73	92.5%	80.6%	86.1%	92.1%
Dry Etch	21	312	88.5%	80.4%	85.1%	91.5%
Implant	9	36	80.2%	70.8%	56.1%	85.6%
Lithography	11	170	86.4%	51.6%	85.6%	96.8%
Planar	6	26	80.0%	90.3%	69.7%	89.4%
Thin Films	11	79	85.2%	71.1%	79.2%	89.2%
Wet Etch	14	99	89.1%	80.5%	77.3%	90.9%
Defectivity Metrology	7	15	96.5%	89.8%	45.5%	78.4%
Litho Metrology	4	45	96.5%	90.0%	87.7%	91.8%
Thin Films Metrology	2	4	96.5%	90.3%	45.0%	72.7%
Total	105	913				

By design, we try to make the lithography area the fab constraint by adjusting the number of tools in preliminary simulation experiments. Consequently, the highest utilization figures above 92% belong to lithography tools. At the other extreme, most tool groups presenting a very low utilization are those where only a single tool is required, but two tools are installed to ensure redundancy.

B. Operating Curve

The behavior of the fab is typically represented by what is denoted as its operating curve, presenting the evolution of CT as a function of the fab load. In our case, in order to unify the results of several products, we show FF values instead of CT values. We present in Figure 2 and Figure 3 the percentiles of the FF curves as a function of the load (ratio of wafer starts and what has been determined as the model capacity, i.e., 10,200 and 10,250 for datasets 1 and 2 respectively). The lowest and highest curves are for the shortest and longest CT respectively, with the other curves ordered one above the other for percentiles from 5% to 95% in an increasing order. Only regular lots are represented, hot lots have a FF close to 1.0. The quantiles shown in Figures 2 and 3 are computed based on a single simulation run with a length of eight years where the first year serves as warmup period. All lot cycle times are collected to derive the order statistics which are required to compute the quantiles. More advanced sequential procedures for estimating quantiles with a given precision such as the zoom in (ZI) procedure [40] are not used in this paper since the importance of the precision is secondary to the main goal of the paper which consists in introducing the models and showing their general behavior.

At 10,000 WSPW (the proposed working point for both models), the median of FF is 1.85 and 1.95, for datasets 1 and 2, respectively, while the maximum reach values of 2.37 and 2.46, respectively. 90% of all lots have a FF between 1.73 and 2.01 for dataset 1, and between 1.79 and 2.13 for dataset 2 corresponding to the 5% and 95% percentiles, respectively.

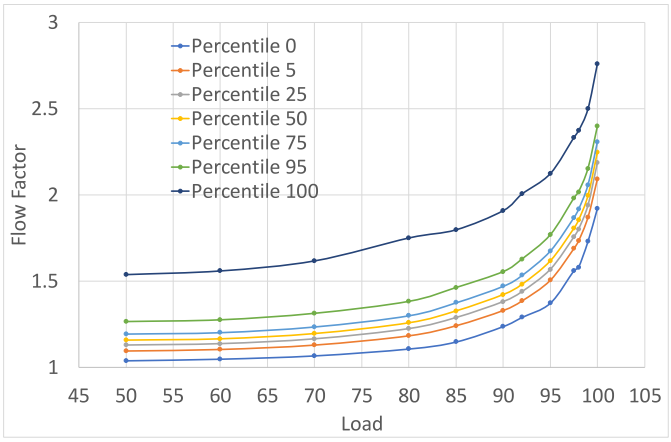


Fig. 2. Operating curves for dataset 1.

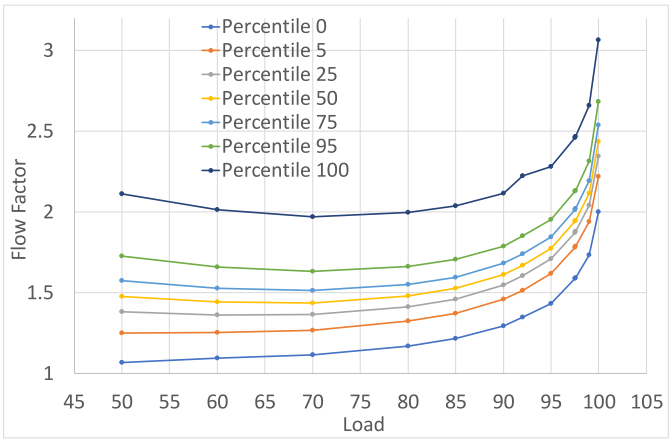


Fig. 3. Operating curves for dataset 2.

Interestingly, we remark that for dataset 2, and for the largest FF values (percentile 75 and above) the curves are first decreasing with the load. This seemingly counter-intuitive phenomenon is due to the fact that for low fab loadings some lots undergo long periods of waiting at diffusion tools for batches to be formed. This does not occur in dataset 1 because it is a high-volume scheme in which the number of lots of each type is five times higher than in dataset 2 (for each load, half of the products are of the same type in dataset 1 and only one tenth in dataset 2). Detailed performance measure values for datasets 1 and 2 can be found in Appendix I of the electronic supplement.

V. CASE STUDY

A. Dispatching Strategies for Production and Engineering Lots

This study aims at demonstrating the usage of models from the testbed to answer questions that require some extension of the models. We will assess the performance of dispatching strategies for production and engineering activities that are already used for the Kayton model [10] and the MIMAC 1

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TABLE V
DESIGN OF EXPERIMENTS

Factor	Level	Count
Dataset	3, 4	2
Dispatching strategy	BASE, EF, CAtE (168h, 84h, 24h), CoT (100, 50, 25, 10)	9
Number of independent simulation replications	20	
Total simulation runs		360

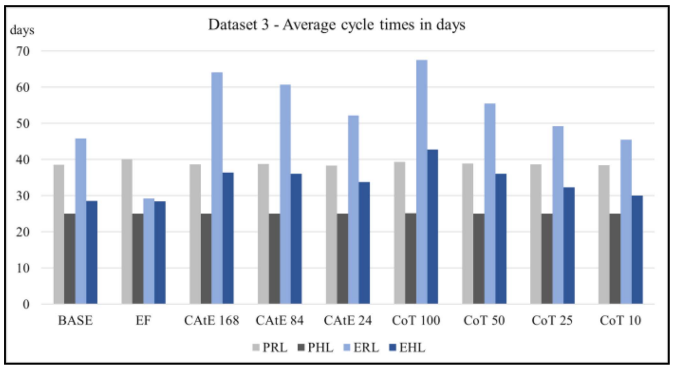


Fig. 4. Average cycle times in days for dataset 3.

dataset [28]. Note that proposing new dispatching strategies is beyond the scope of this study.

A basic dispatching strategy, abbreviated by BASE, is used for all tool groups in datasets 3 and 4. Production hot lots (PHL) and engineering hot lots (EHL) have a higher priority than production-related regular lots (PRL) and engineering-related regular lots (ERL), i.e., hot lots are always preferred to regular lots. Given lots with the same priority, FCFS for dataset 3 and CR for dataset 4 are applied. The priorities for EHL, PHL, ERL, and PRL are 20, 20, 10, and 10, respectively. Moreover, integrated dispatching strategies similar to the ones from [10] for both production lots (PLs) and engineering lots (ELs) are applied. PLs and ELs consist of hot and regular production and engineering lots, respectively. The first additional strategy is called engineering first (EF), where ELs have a higher priority than PLs, i.e., if lots of type EL are available they are always preferred. The priorities for EHL, PHL, ERL, and PRL are chosen as 25, 20, 15, and 10, respectively. The EF strategy is applied to all tool groups.

The capacity allocation to engineering (CAtE) strategy only allows engineering activities within so-called engineering intervals with the exception that if no ELs are available within the engineering interval, production activities are allowed and vice versa. The lengths of the engineering and production intervals have to be chosen. Production and engineering intervals are repeated in a consecutive manner, i.e., it is enough to specify pairs (lp, le) where lp is the length of the production interval and le the length of the engineering interval. The settings (151.2, 16.8), (75.6, 8.4), and (21.6, 2.4) in hours are used for dataset 3, whereas (134.6, 33.4), (67.2, 16.8), and (19.2, 4.8) are the corresponding settings for the simulation experiments with dataset 4. This leads to a periodicity of one week, half a week, and a single day.

A tool group-specific number of lots, a so-called trigger limit, must be prescribed for the changeover trigger (CoT) strategy. If the number of ELs waiting in front of a particular tool group reaches or exceeds the trigger limit, this number of ELs will be processed. Trigger limits of 100, 50, 25, and 10 lots are applied for both datasets in the experiments.

Both CAtE and CoT are only used for dispatching on the LithoTrack_FE_95 and LithoTrack_FE_115 tool groups which are formed by steppers, typical bottlenecks in wafer fabs. For the remaining tool groups, the BASE strategy is applied. The design of experiments is summarized in Table V.

A simulation horizon of two years including a warm-up phase of one year which is excluded from the statistics is used. We report the global performance measure values throughput (TH), i.e., the number of completed lots within the simulation horizon, the average cycle time (ACT) in days, and the percentage of lots finished on time (%ONTIME). Average values of all performance measure values are provided based on 20 independent simulation replications.

The commercial simulation engine AutoSched AP 11.3.0 is applied. The CAtE and CoT strategies are coded in the C++ programming language applying customization features of the AutoSched AP software.

B. Performance Assessment

Detailed simulation results are only provided in Appendix II of the electronic supplement due to space limitations. The ACT values for datasets 3 and 4 are shown in Figures 4 and 5 where the results are grouped according to the EHL, PHL, ERL, and PRL lot types.

First, we see from Tables AII-1 and AII-3 in the supplementary material that the ACT values of the different ELs are less than the ones of the corresponding PLs since the number of wafers is much smaller for the ELs. We observe from the tables and the Figures 3 and 4 that the performance measure values of the ELs are the best under the EF strategy. However, at the same time the performance measure values of the PLs are the worst compared to all other strategies. The Tables AII-1 - AII-4 in the supplementary material show that a long production interval for the CAtE policy results in less TH and larger ACT values for ELs and less ELs completed on time due to long waiting times in front of the tool groups.

The same behavior is observed for large values of the trigger limit. We see from Table AII-1, in the supplementary material that in these situations the ACT values of the PL and the percentage of PLs finished on time are worse too. But if the CR rule is used as for dataset 4, this deterioration of the performance values for PLs cannot be observed (see Tables AII-3, AII-4, in the supplementary material and Figure 4) since the effect is more balanced. This is because the CAtE and CoT strategies are only applied on two stepper tool groups.

Moreover, we can see that the deterioration for ELs is reduced, i.e., the TH and the ACT values are very similar for ERLs for different parameterizations of the CAtE and CoT

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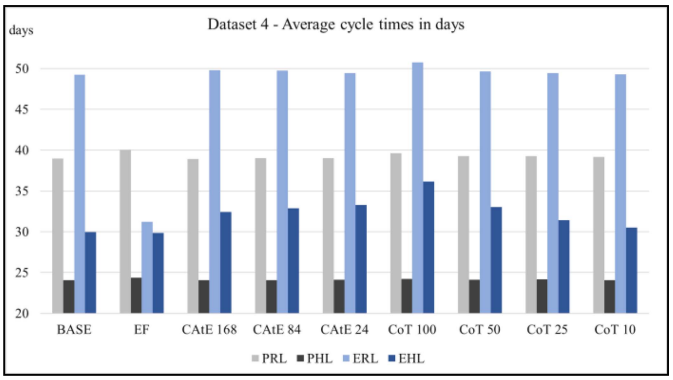


Fig. 5. Average cycle times in days for dataset 4.

strategies. However, to achieve short CT values as for the EHLs long production intervals or high trigger limits are more difficult to compensate.

VI. Accessing and Using the Datasets

The testbed was created and tested using the AutoSched AP simulation software. The four simulation models with the discussed features are publicly available at [16]. We provide access to the testbed models on the following levels:

1. The four datasets of the testbed are represented as a set of EXCEL sheets in a generic, i.e., simulation tool-independent manner (files SMT_20-20_Model_Data_*.xlsx in the folder General_Data of the testbed).
2. We provide a detailed documentation of the testbed for users of Autosched AP (file SMT_2020_AutoSched_AP_documentation.pdf in the AutoSched folder of the testbed). The file contains the technical information necessary to navigate and use the data files of the simulation models. The main features of the different datasets are described and their implementation using AutoSched AP are explained. Applicants of the testbed can use the simulation models directly in AutoSched AP. They are located in the AutoSched folder of the testbed.
3. It is also possible to use the datasets for other discrete-event simulation tools. For instance, the simulation engine AnyLogic is popular in semiconductor manufacturing [41]. To make such attempts easier we also provide a simulation tool-independent documentation of the datasets by means of the file SMT_2020_generic_format.pdf which is again located in the General_Data folder of the testbed [16].

The following procedure is recommended to use the datasets:

1. In a first step, the model has to be implemented using the simulation tool of choice. Alternatively, the provided simulation models can be directly used if the AutoSched AP software is available.
2. Simulation runs have then to be carried out based on the simulation models from Step 1. Statistics have to be gathered for TH and CT for all the models, whereas the number of ontime lots has to be additionally collected for the LV/HM models. Simulation results can be found in the electronic supplement for a direct comparison.
3. If the values of the performance measures of interest obtained in Step 2 are close to the ones from the supplement, the simulation models can be used to answere specific questions. This might require to add additional functionality to the simulation models or to couple the simulation models with planning and control algorithms. In the latter situation, the infrastructure sketched in [15], [42] can be applied.

VII. Conclusion

Concluding a two-year long development effort, we present the semiconductor manufacturing researchers and practitioners the new simulation testbed SMT2020 which aims at mimicking the behavior of modern fabs. During this development phase, the main ideas and preliminary results were presented in [8], [9] to the community who was asked to provide feedback. We carefully incorporated the required features and validated the result to ensure the relevance of our models. The core of SMT2020 is composed of a set of four models, namely datasets 1 and 2 which provide the foundation of the testbed by representing a HV/LM fab and a LV/HM fab respectively and datasets 3 and 4 which add engineering lots to datasets 1 and 2, respectively. SMT2020 is available under [16], and we encourage researchers to use it when issues relevant to modern fabs are to be studied. The characteristics of the simulation models are reviewed herein and we presented important performance measure values to allow comparisons with future bodies of work.

We demonstrated the usage of the testbed datasets by performing a simulation study where we assessed the performance of integrated dispatching strategies for production and engineering activities. Datasets 3 and 4 were extended by adding customized dispatching strategies. Related performance measure values are reported.

There are several directions for future research. Consistent with the idea of tackling managerial issues related to features that emerged in modern fabs, our planned next steps are to tackle the issue of CQT constraints and how to reduce their impact on operation. As a second research avenue, we are interested in extending the integrated planning formulations from [43] for production and engineering activities towards datasets 3 and 4. It is also desirable to include some of the SMT2020 models as nodes in the semiconductor supply chain testbed proposed in [15].

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